Certified Grid-Side Behavior of an Uninterruptible Power Supply for AI Training Loads

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Abstract: **Synchronized AI training jobs swing the power of a whole campus between its floor and its peak almost as a step, and grid operators now ask such loads for guarantees that these swings stay off the grid. We study a battery and inverter placed between the campus and the grid, run by a deliberately simple grid-side law: a rate-clipped first-order low-pass of the load plus a saturated loop that restores a *settled* state of charge. For every measurable load profile with values in a fixed interval, with steps allowed, we prove that the grid power ramps no faster than a prescribed limit, that the battery stays inside its power rating and energy window, and that the point-of-common-coupling voltage stays in a band and never steps. The battery claim rests on a polytopic robust invariant set whose invariance condition is verified by interval branch-and-bound with outward rounding, and we state precisely why that check implies the claim. For a public, assumed-parameter campus of 60 to $100\,\mathrm{MW}$ and a ramp limit of $10\,\mathrm{MW/min}$, the certificate needs a $40.28\,\mathrm{MW}$, $2.013\,\mathrm{MWh}$ battery, within $1.43\times$ in energy of a lower bound that we prove holds for *any* causal controller. Ablations show that a plain ramp-rate limiter drains the same battery on admissible burst trains, and that restoring the raw state of charge instead of the settled one costs $2.83\times$ more energy.**

Index Terms: **Battery energy storage, data centers, robust invariance, interval arithmetic, ramp-rate limits, verification.**

I. Introduction

Large AI training jobs run in lockstep: thousands of accelerators compute together, then communicate or checkpoint together, so the campus power swings with every training iteration, and the swings grow with job size [1]. Grid operators have started to treat such loads as a reliability topic in their own right. The NERC Large Loads Task Force lists rapid load swings among the risks to frequency and voltage regulation [2]; ERCOT has surveyed large electronic loads of $75\,\mathrm{MW}$ and above about their ride-through behavior [3]; and interconnection practice increasingly attaches ramp-rate conditions to such loads [4]. One industrial response is an "AI UPS": a medium-voltage battery and inverter that sits between campus and grid and acts as a firewall in both directions.

A firewall is only useful if its behavior can be guaranteed for *every* load the campus might produce, not only for the profiles that were simulated. This paper gives such a guarantee for a simple, implementable controller, in the style of control-theoretic stability results for grid-forming converters [5]: a model with numbered equations, formal assumptions, and theorems with proofs. The one step that is not closed form, robust invariance of a battery state set under a saturated nonlinear vector field, is discharged by a computer check whose soundness we argue explicitly.

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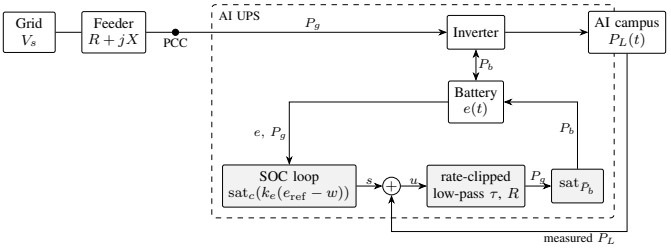


Fig. 1. Grid, feeder, point of common coupling (PCC), AI UPS and campus. Power flows on the top row; the controller (shaded) sets the grid-side power P_g from the measured load P_L, the stored energy e and P_g itself; the battery supplies the difference $P_b = P_L - P_g$.

Contributions. (i) A grid-side law that restores the *settled* state of charge $w = e + \tau(P_g - P_{\mathrm{mid}})/3600$ rather than the raw state of charge e (Section II). (ii) A band lemma (Lemma 2) that turns robust invariance of a polytope under measurable inputs into a finite set of inequalities on boxes, and a sound interval branch-and-bound that proves them (Proposition 1). (iii) Ramp, battery and voltage guarantees for every admissible profile (Theorem 1, Corollary 1), and a controller-independent lower bound on battery size (Proposition 2) against which the certified sizes are measured. (iv) A battery-size versus ramp-limit curve in which every point is certified.

II. System Model

Fig. 1 shows the configuration. Powers are in MW, energies in MWh, time in seconds.

Load. The campus draws

$$P_L(t) \in \mathcal{D} := [P_{\min}, P_{\max}], \qquad \Delta P := P_{\max} - P_{\min}, \quad (1)$$

with no bound on $|dP_L/dt|$. We use $P_{\min} = 60$, $P_{\max} = 100$, so $\Delta P = 40$, and $P_{\mathrm{mid}} = 80$.

Battery power. By power balance at the UPS, with a power rating $\bar{P}_b$,

$$P_b = \mathrm{sat}_{\bar{P}_b}(P_L - P_g), \qquad P_{\mathrm{grid}} = P_L - P_b, \qquad (2)$$

where $\mathrm{sat}_a(x) = \mathrm{clip}(x, -a, a)$ and $P_b > 0$ means discharge.

State of charge. With one-way efficiency $\eta \in (0, 1]$,

$$\dot{e} = -\tfrac{1}{3600}\,\varphi(P_b), \qquad \varphi(p) = \begin{cases} p/\eta, & p \geq 0, \\ \eta p, & p < 0, \end{cases} \qquad (3)$$

with e measured from a reference $e_{\mathrm{ref}} = 0$. We take $\eta = 0.95$ (assumed).

Controller. Let $\tau > 0$, $R > 0$, $c > 0$, $k_e > 0$ and define the settled state of charge

$$w := e + \tfrac{\tau}{3600}(P_g - P_{\mathrm{mid}}). \qquad (4)$$

The grid-side law is

$$\begin{aligned} \dot{P}_g &= \mathrm{clip}\Big(\frac{P_L + s - P_g}{\tau}, -R, R\Big), \\ s &= \mathrm{sat}_c\big(k_e(e_{\mathrm{ref}} - w)\big). \end{aligned} \tag{5}$$

Without the clip and without s, (5) is a first-order low-pass of P_L; the clip enforces the ramp limit by construction; s slowly returns the battery toward its reference. The quantity w is the energy the battery will hold once P_g has settled at P_{mid} (Lemma 1(c)); regulating w rather than e stops the SOC loop from fighting the filter and shrinks the certified energy window by a factor of 2.83 (Section IV). We parametrize $\tau = \alpha\,\Delta P/R$ and $k_e = c/(\kappa E_0)$ with $E_0 := \tau\Delta P/3600$.

Feeder voltage. The PCC is fed from a stiff source V_s through $Z = r + jx$ (per unit, no shunt). For a grid-side draw $P + jQ$ (per unit) the exact PCC magnitude on the normal operating branch is

$$\begin{aligned} V^2 &= \tfrac{1}{2}\Big(A + \sqrt{A^2 - 4|Z|^2(P^2+Q^2)}\Big), \\ A &= V_s^2 - 2(rP + xQ), \end{aligned} \tag{6}$$

and its linearization is

$$V \approx V_s - (rP + xQ)/V_s. \tag{7}$$

We use a 69 kV, 10 km line with $0.05 + j0.35\,\Omega$/km on a 100 MVA base ($r = 0.0105$, $x = 0.0735$), $V_s = 1$, the UPS at unity power factor, and, without the UPS, the campus at power factor 0.95 lagging (all assumed). An AC power flow in pandapower [12] on the same feeder agrees with (6) to within 6.7×10^{-12} pu at 41 operating points.

The closed loop (2) to (5) has state $x = (P_g, e)$ and input P_L. We write it $\dot{x} = f(x, P_L)$.

Assumption 1. *$P_L : [0,\infty) \to \mathcal{D}$ is Lebesgue measurable.*

Assumption 2. *The inverter tracks its power reference exactly (no fast current or PLL dynamics), the battery parameters are constant, the battery can hold every energy in $[e_{\mathrm{lo}}, e_{\mathrm{hi}}]$ given below, and the source behind the feeder is stiff.*

Under Assumption 1, f is globally Lipschitz in x (it is built from affine maps, clip and the piecewise-linear φ) and measurable and bounded in t, so for every $x(0)$ there is a unique absolutely continuous (Carathéodory) solution on $[0,\infty)$.

Definition 1. *A set $\mathcal{S} \subset \mathbb{R}^2$ is* robustly forward invariant *if every solution with $x(0) \in \mathcal{S}$ and any P_L satisfying Assumption 1 has $x(t) \in \mathcal{S}$ for all $t \geq 0$.*

Problem 1. *Given R, find controller parameters, a battery rating $\bar{P}_b$, an energy window $[e_{\mathrm{lo}}, e_{\mathrm{hi}}]$ of smallest width, and a set $\mathcal{S}$ of initial states such that for every admissible P_L and every $x(0) \in \mathcal{S}$: (a) $|P_{\mathrm{grid}}(t_2) - P_{\mathrm{grid}}(t_1)| \leq R|t_2 - t_1|$; (b) $|P_b(t)| \leq \bar{P}_b$ and $e(t) \in [e_{\mathrm{lo}}, e_{\mathrm{hi}}]$; (c) the PCC voltage stays in $[1 - 0.05, 1 + 0.05]$ and is Lipschitz in time.*

III. Main Results

A. The unsaturated filter in closed form

Lemma 1. *Consider (5) with $s \equiv 0$ and no clip, i.e. $\dot{P}_g = (P_L - P_g)/\tau$. For every admissible P_L: (a) if $P_g(0) \in \mathcal{D}$ then $P_g(t) \in \mathcal{D}$ for all $t \geq 0$; (b) then $|\dot{P}_g(t)| \leq \Delta P/\tau$ for almost every t; (c) if moreover $\eta = 1$ and $|P_L - P_g| \leq \bar{P}_b$, then $w(t) = w(0)$ for all t.*

Proof. (a) At $P_g = P_{\max}$, $\dot{P}_g = (P_L - P_{\max})/\tau \leq 0$, and at $P_g = P_{\min}$, $\dot{P}_g \geq 0$; for the scalar absolutely continuous P_g this keeps $P_g \in \mathcal{D}$ (if $P_g(t_1) > P_{\max}$, let t_0 be the last time before t_1 with $P_g = P_{\max}$; on $(t_0, t_1]$ $\dot{P}_g < 0$ a.e., a contradiction). (b) Both P_L and P_g lie in $\mathcal{D}$, so $|P_L - P_g| \leq \Delta P$. (c) $\dot{w} = \dot{e} + \frac{\tau}{3600}\dot{P}_g = \frac{1}{3600}\big(-(P_L - P_g) + (P_L - P_g)\big) = 0$. □

Lemma 1(b) is the closed-form ramp bound: $\tau \geq \Delta P/R$ suffices. Two things break the closed form. Losses make $\dot{w} = -\frac{1}{3600}(\varphi(P_b) - P_b) \leq 0$, so w drifts down forever on any oscillating load and the battery eventually empties; the SOC loop s is needed to stop that drift, and once it is present P_g can leave $\mathcal{D}$. The clip lets $\tau < \Delta P/R$, which shrinks the battery, but then w is no longer conserved during clipped motion. Battery invariance for the full law (5) is what we certify next.

B. A band lemma for polytopes

Let $z(x) = \big((P_g - P_{\mathrm{mid}})/\Delta P,\ e/E_0\big)$ and, for unit normals n_k and offsets b_k, $k = 1, \ldots, K$,

$$h_k(x) = n_k^\top z(x) - b_k, \quad \mathcal{S} = \{x : B(x) := \max_k h_k(x) \leq 0\}. \tag{8}$$

B is a max of affine functions, i.e. a one-hidden-layer ReLU function, and $\mathcal{S}$ is a convex polygon. For $\varepsilon > 0$ define the outer bands $\mathcal{B}_k^\varepsilon = \{x : 0 \leq h_k(x) \leq \varepsilon,\ h_j(x) \leq \varepsilon\ \forall j \neq k\}$.

Lemma 2 (band lemma)**.** *Suppose that for every k,*

$$\nabla h_k^\top f(x,p) \leq 0 \quad \forall x \in \mathcal{B}_k^\varepsilon,\ p \in \mathcal{D}. \tag{9}$$

Then $\mathcal{S}$ is robustly forward invariant.

Proof. Take an admissible P_L, a solution with $x(0) \in \mathcal{S}$, and suppose $x(t_1) \notin \mathcal{S}$ for some $t_1 > 0$. Let $t^* = \sup\{t \leq t_1 : x(t) \in \mathcal{S}\}$; $\mathcal{S}$ is closed, so $x(t^*) \in \mathcal{S}$ and $t^* < t_1$. By continuity there is $\delta > 0$ with $h_j(x(t)) \leq \varepsilon$ for all j and $t \in [t^*, t^* + \delta]$, and since $x(t) \notin \mathcal{S}$ for $t \in (t^*, t_1]$ there are $t_2 \in (t^*, t^* + \delta]$ and k with $h_k(x(t_2)) > 0$. Let $t_3 = \sup\{t \leq t_2 : h_k(x(t)) \leq 0\} \geq t^*$, so $h_k(x(t_3)) = 0$ and $h_k(x(t)) \in (0, \varepsilon]$ on $(t_3, t_2]$. There $x(t) \in \mathcal{B}_k^\varepsilon$, so by (9) $\frac{d}{dt}h_k(x(t)) = \nabla h_k^\top f(x(t), P_L(t)) \leq 0$ for almost every t. Since $h_k \circ x$ is absolutely continuous, $h_k(x(t_2)) \leq h_k(x(t_3)) = 0$, a contradiction. □

Condition (9) involves no time and no input signal, only the box $\mathcal{B}_k^\varepsilon \times \mathcal{D}$. That is what makes it checkable.

C. Sound verification

For each k the branch-and-bound (BaB) starts from one box in (P_g, e, P_L) that contains $\mathcal{B}_k^\varepsilon \times \mathcal{D}$ (the bounding box of $\{B \leq \varepsilon\}$ padded by 0.05 in normalized units), and repeatedly bisects boxes along their widest normalized side. On a box $\mathbf{X}$ it evaluates interval enclosures $[h_j](\mathbf{X})$ and $[\nabla h_k^\top f](\mathbf{X})$, and discards $\mathbf{X}$ only if (i) $[h_k]$ misses $[0, \varepsilon]$, or some $[h_j]$,

$j \neq k$, lies above ε (then $\mathbf{X} \cap \mathcal{B}_k^{\varepsilon} = \emptyset$), or (ii) the upper end of $[\nabla h_k^{\top} f](\mathbf{X})$ is ≤ 0. It reports failure if it finds a box midpoint in the band with positive derivative, or reaches depth 60 or its box budget.

Soundness. Bisection keeps the live boxes plus the discarded ones a cover of the initial box. Every interval operation (addition, multiplication, division by a constant, square root) rounds its lower end down and its upper end up by one unit in the last place; IEEE 754 arithmetic is correctly rounded, so each computed interval contains the exact real result [11]. clip, sat and φ are monotone, so their images are evaluated exactly at the endpoints. Hence an interval enclosure is a superset of the true range, and each discard rule is implied by a true statement about every point of the box. If the BaB terminates with all boxes discarded, (9) holds for face k.

The vector field in (9) is evaluated with an unsaturated battery, $P_b = P_L - P_g$. The same BaB proves $|P_L - P_g| \leq \bar{P}_b$ on $\{B \leq \varepsilon\} \times \mathcal{D}$, where both fields agree, so (9) holds for the true f. It also proves the bounds $e_{\mathrm{lo}} \leq e \leq e_{\mathrm{hi}}$ and $P_g^{\mathrm{lo}} \leq P_g \leq P_g^{\mathrm{hi}}$ on $\mathcal{S}$. Sampling is used only to propose the offsets b_k (faces whose sampled derivative is not strictly negative are pushed out); it plays no role in the proof.

Proposition 1 (battery invariance)**.** *Let $R = 10\,$MW/min, $\tau = 180\,$s, $c = 10\,$MW, $k_e = 100.00\,$MW/MWh, $\eta = 0.95$, and let $\mathcal{S}$ be the 32-face polygon* (8) *stored in* `results/certificate.json`*, with $\varepsilon = 0.002$. The BaB proves* (9) *for all faces, using 27,230 boxes, maximum depth 40, in 1.4 s. Consequently, for every admissible P_L and every $x(0) \in \mathcal{S}$: $x(t) \in \mathcal{S}$, $|P_L(t) - P_g(t)| \leq 40.28\,$MW, $e(t) \in [-1.006, 1.006]\,$MWh and $P_g(t) \in [59.81, 100.19]\,$MW for all $t \geq 0$. With $\bar{P}_b \geq 40.28\,$MW the battery power limit never binds, $|P_b(t)| \leq 40.28\,$MW, and the usable energy needed is $2.013\,$MWh.*

Proof. The BaB terminated with every box discarded for every face, so (9) holds by the soundness argument above, and Lemma 2 gives invariance. The remaining bounds were proved by the same BaB on $\mathcal{S}$ (or on $\{B \leq \varepsilon\} \supset \mathcal{S}$ for $|P_L - P_g|$) and hold along the trajectory because it stays in $\mathcal{S}$. □

D. Ramp and voltage guarantees

Theorem 1 (ramp limit)**.** *Under the hypotheses of Proposition 1 with $\bar{P}_b \geq 40.28\,$MW and an energy window containing $[-1.006, 1.006]$, for every admissible P_L and every $x(0) \in \mathcal{S}$, $P_{\mathrm{grid}}(t) = P_g(t)$ and*

$$|P_{\mathrm{grid}}(t_2) - P_{\mathrm{grid}}(t_1)| \leq R\,|t_2 - t_1| \qquad \forall t_1, t_2 \geq 0.$$

Proof. By Proposition 1, $|P_L - P_g| \leq \bar{P}_b$ along the trajectory, so the saturation in (2) is inactive and $P_{\mathrm{grid}} = P_L - P_b = P_g$; the energy limits are never reached, so no protection action changes P_b. P_g is absolutely continuous with $|\dot{P}_g| \leq R$ a.e. by the clip in (5), hence R-Lipschitz. □

Without the battery bound the clip alone proves nothing: if the battery saturates or empties, the grid absorbs the difference $P_L - P_g - P_b$ instantly and sees the load step (Section IV).

Corollary 1 (voltage band)**.** *Under Theorem 1, with the UPS at unity power factor and the feeder of Section II, the PCC voltage* (6) *satisfies $V(t) \in [0.9865, 0.9927]\,$pu, so $|V - 1| \leq 1.35\%$, for all t; and it changes by at most 0.165% per minute. The linearization* (7) *differs from* (6) *by at most $2.94 \times 10^{-3}\,$pu over the certified range of P_g.*

Proof. V is a continuous function of $P_g \in [59.81, 100.19]$ (Proposition 1). Covering that interval by 4000 subintervals and evaluating (6), its derivative and $V - V_{\mathrm{lin}}$ in outward-rounded interval arithmetic gives the stated range, a bound L_V on $|dV/dP_g|$, and the linearization error; then $|V(t_2) - V(t_1)| \leq L_V |P_g(t_2) - P_g(t_1)| \leq L_V R |t_2 - t_1|$ by Theorem 1. □

E. How small can the battery be?

Proposition 2 (lower bound for any controller)**.** *Let any causal controller choose P_g with $|P_g(t_2) - P_g(t_1)| \leq R|t_2 - t_1|$, and suppose that for every admissible P_L it keeps $|P_L - P_g| \leq \bar{P}_b$ and $e \in [e_{\mathrm{lo}}, e_{\mathrm{hi}}]$ under* (3)*. Then*

$$\bar{P}_b \geq \Delta P, \qquad e_{\mathrm{hi}} - e_{\mathrm{lo}} \geq \frac{\Delta P^2}{2\eta R \cdot 3600}\ \text{MWh}.$$

Proof. Fix $\delta \in (0, \Delta P)$ and the profile $P_L \equiv P_{\min}$. If $P_g(t) > P_{\min} + \delta$ for all t, then $\dot{e} = \eta(P_g - P_{\min})/3600 > \eta\delta/3600$ and e is unbounded, which is excluded. So there is T with $P_g(T) \leq P_{\min} + \delta$. Now take the profile equal to $P_{\min}$ on $[0, T)$ and $P_{\max}$ afterwards. By causality P_g is unchanged on $[0, T]$, and for $t \geq T$, $P_g(t) \leq P_{\min} + \delta + R(t - T)$. Thus $P_b(T^+) \geq \Delta P - \delta$, and over $[T, T + (\Delta P - \delta)/R]$ the battery delivers at least $(\Delta P - \delta)^2/(2R)$ MW s, drawing at least $(\Delta P - \delta)^2/(2\eta R \cdot 3600)$ MWh from storage. Since $e(T) \leq e_{\mathrm{hi}}$ and e stays above e_{lo}, the window is at least that large. Let $\delta \to 0$. □

At $R = 10\,$MW/min the bound is $40\,$MW and $1.404\,$MWh. The certified design uses $1.007\times$ the power and $1.43\times$ the energy.

Remark 1 (scaling in R). With $\tau = \alpha \Delta P/R$ and $k_e = c/(\kappa E_0)$, the change of variables z in (8) and time $t' = tR/\Delta P$ makes the closed loop independent of R. So the same normalized polygon is invariant for every R, the certified energy window scales as $1/R$, and $\bar{P}_b$ does not depend on R, exactly like the lower bound of Proposition 2. We nevertheless re-run the BaB at each R of Table I, since the floating-point problem differs.

IV. Numerical Results

All numbers below are produced by `run_all.py` and read from `results/*.json`.

Design search. At $R = 10\,$MW/min we searched $\alpha \in \{0.5, 0.6, 0.75, 0.9, 1\}$, $c \in \{3, 6, 10\}\,$MW and $\kappa \in \{0.5, 0.15, 0.05\}$, 45 designs, of which 29 were certified; the others had no invariant 32-face candidate (their offsets diverged), which is a failure of the search, not a proof that no invariant set exists. The design with the smallest certified window has $\alpha = 0.75$, $c = 10\,$MW, $\tau = 180\,$s.

TABLE I
CERTIFIED BATTERY SIZE VERSUS RAMP LIMIT

R MW/min	α	c MW	τ s	$\bar{P}_b$ MW	window MWh	bound MWh	ratio	BaB boxes
2	0.75	10	900	40.28	10.064	7.018	1.43	27,230
5	0.75	10	360	40.28	4.026	2.807	1.43	27,230
10	0.75	10	180	40.28	2.013	1.404	1.43	27,230
20	0.75	10	90	40.28	1.006	0.702	1.43	27,230
40	0.75	10	45	40.28	0.503	0.351	1.43	27,230

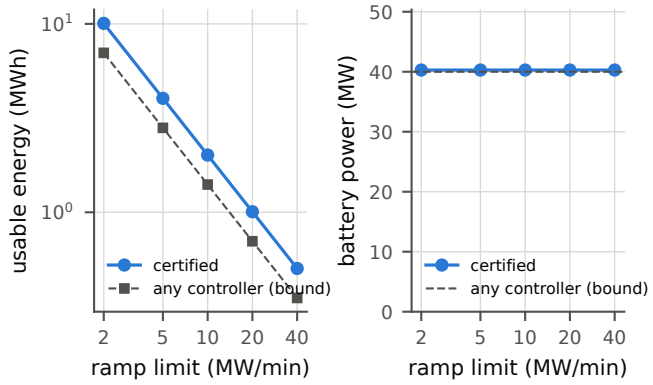


Fig. 2. Battery size versus ramp limit. Left: certified usable energy and the lower bound of Proposition 2 (both scale as $1/R$). Right: certified power rating and the bound ΔP.

Sizing curve. Table I and Fig. 2 give the certified battery against the ramp limit, each row re-certified by BaB at its own R, together with the bound of Proposition 2.

Invariant set. Fig. 3 shows $\mathcal{S}$ with 210 simulated profiles from the three load families (square bursts with period 2 to 30 s, duty 0.2 to 0.8, depth 10 to 40%; checkpoint dips to idle; random walks) and a greedy adversary that, at each step, picks the load value that pushes the active face outward fastest. On all simulated trajectories $\max B = -0.002$ (families), -0.002 (330 square waves with periods up to 960 s) and -0.002 (adversary), all inside $\mathcal{S}$ as the proposition requires. The adversary reaches stored energies in $[-1.000, 1.000]$ MWh against the certified $[-1.006, 1.006]$ MWh, so in the energy direction the polygon is close to tight for this design. The dashed outline is the set certified for the plain loop that restores e instead of w, with the same (α, c, κ): its window is 5.700 MWh, 2.83× larger.

Worst-case profiles. Over the square-wave grid, the deepest discharge came from period 960 s, duty 0.95, depth 10%. Fig. 4 shows the three extreme profiles. The grid power is smooth in every case; the largest simulated battery power was 39.96 MW against the certified 40.28 MW.

Voltage. Fig. 5 compares the PCC voltage on a checkpoint profile. Without the UPS the voltage stays in $[0.9613, 0.9778]$ pu, inside the 5% band, but a full load step moves it by 1.65% instantly. With the UPS it stays in $[0.9865, 0.9927]$ pu and moves at most 0.165% per minute (Corollary 1). On this assumed stiff feeder the gain is the removal of steps, not the band itself. The linearization error over the no-UPS range is at most 4.06×10^{-3} pu; against pandapower the linear model is off by at most 4.07×10^{-3} pu.

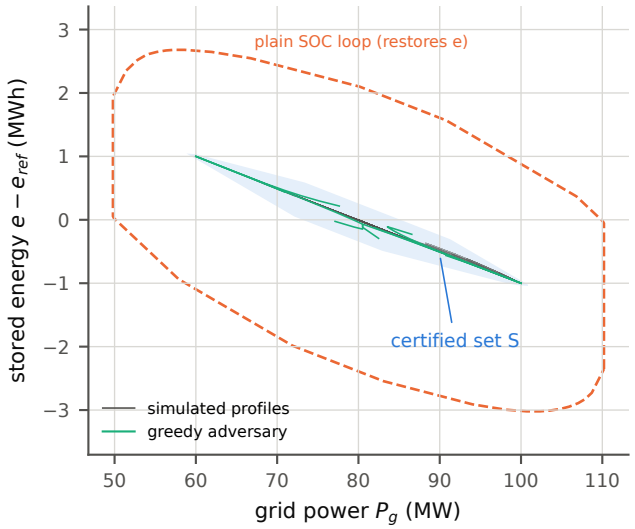


Fig. 3. Certified invariant set $\mathcal{S}$ in the (P_g, e) plane (filled), simulated trajectories (gray), greedy adversary (green), and the larger set certified for the plain SOC loop (dashed).

Ablations. Without a UPS the grid sees P_L itself: a 40 MW step within one 0.1 s simulation sample, 400 MW/s at that resolution and unbounded in the limit, against $R = 0.1667$ MW/s. A classic ramp-rate limiter ($\dot{P}_g = \mathrm{clip}((P_L - P_g)/1\,\mathrm{s}, -R, R)$, no SOC loop) given the same certified battery drained it on 60 of 140 burst trains from the brief's family within 2 h; the fastest was period 5 s, duty 0.3, depth 20%, empty after 17.2 min, after which the grid sees ramps of 199.9 MW/s (Fig. 6). The reason is the asymmetry of short bursts: during a short high phase the limiter cannot catch up, so the battery supplies most of every burst and recovers little in between. The certified filter with its SOC loop removed drained the battery on 7 of the same profiles, while the certified controller drained it on 0, with minimum stored energy −0.931 MWh.

V. RELATED WORK

Stability guarantees for grid-forming converters are typically given as theorems on a stated model, as in hybrid angle control [5]; for the broader low-inertia context see [6]. Our object is simpler (a power-reference law, not a converter) but our guarantee is for every input signal in a set, so the tool is set invariance [7], [8] rather than Lyapunov stability. Barrier certificates [9], [10] prove safety with a function whose sublevel set is invariant; our B is such a function, in max-of-affine form, verified by interval analysis [11] instead of sum-of-squares programming. The load side is documented in [1], which characterizes power swings of large training jobs and discusses mitigations framed by utility specifications such as ramp-rate limits. The grid side is in the NERC white paper [2], the ERCOT survey [3], and a survey of interconnection practice [4].

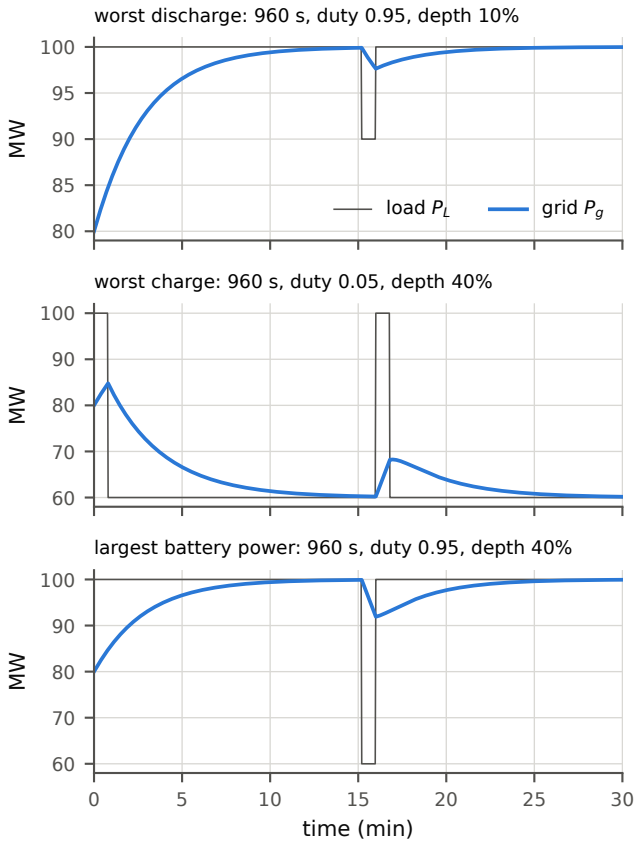


Fig. 4. Load (gray) and grid power (blue) on the three extreme square-wave profiles: deepest discharge, highest charge and largest battery power.

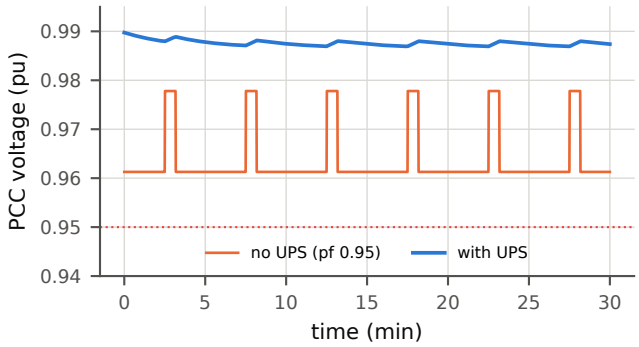


Fig. 5. PCC voltage without the UPS (campus at power factor 0.95) and with the certified UPS, checkpoint profile with a 40 s dip to idle every 300 s. Dotted: lower edge of the 5% band.

VI. Limitations

This is a public, assumed-parameter model. None of the parameters come from ON.energy hardware or data, and the controller is ours, not a vendor's. The proof covers the averaged power model only: no inverter current or PLL dynamics, no switching, no faults, no ride-through events, no protection, no communication delay or measurement noise on P_L, no battery aging, temperature or state-dependent efficiency. The load set allows any measurable profile in $[60, 100]$ MW, but

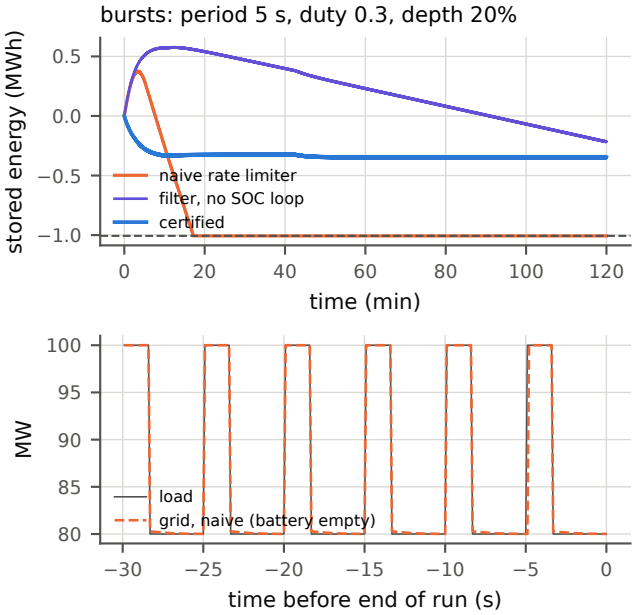


Fig. 6. Ablation on an admissible burst train. Top: stored energy for the naive rate limiter, the filter without SOC loop, and the certified controller; dashed: certified lower energy limit. Bottom: last 30 s of load and grid power for the naive limiter, after its battery is empty: the grid sees every load step.

not excursions outside it (for example a full trip to zero). The feeder is a stiff source behind one line; real PCC voltage depends on the rest of the network and on the campus's own reactive power. The certified set is a 32-face polygon found by a heuristic search over a small design grid, so the sizes are upper bounds for this controller family, and the gap to the lower bound of Proposition 2 may be partly due to the template rather than the controller. Finally, the BaB proof is only as trustworthy as its implementation; the code is small and tested, but it has not been formally verified.

VII. Conclusion

A low-pass with a rate clip and a saturated loop on the settled state of charge is enough to give a provable firewall between an AI campus and the grid: bounded ramp, bounded battery power and energy, and a step-free PCC voltage, for every admissible load profile. The battery it needs is within a small factor of what any controller needs, and the certificate makes clear which piece buys which property.

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